Your Photo Gallery, but Smarter

A Local-First Semantic Image Search System That Runs on Your Laptop

Daniele Giunta

AI Engineer @ Elis Innovation Hub

https://www.linkedin.com/in/daniele-g-dr16/

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I have 40,000 photos.
I can't find any of them.

CTRL+F doesn't work on JPEGs.

What We'll See Today

Index

A library of your own photos, embedded into a single vector space and stored in FAISS on disk.

Search

Natural-language queries – or another photo as the query itself.

Cluster

Unsupervised groups that look like the albums you never made – and labels you never wrote.

Why Local-First?

	Cloud Apps (e.g. Google Photos)	Local-First Apps
Where data lives	🔴 Company servers	🟢 Your device
Privacy	🔴 Company-controlled	🟢 Fully user-owned
Cost model	🔴 Ongoing subscription	🟢 One-time / minimal
Internet dependency	🔴 Required	🟢 None
Setup effort	🟢 Easy	🟢 Simple
Customisable	🔴 Fixed features	🟢 Flexible by design

CLIP – The Text/Image Embedder

«A dog sitting on a chair»

CLIP

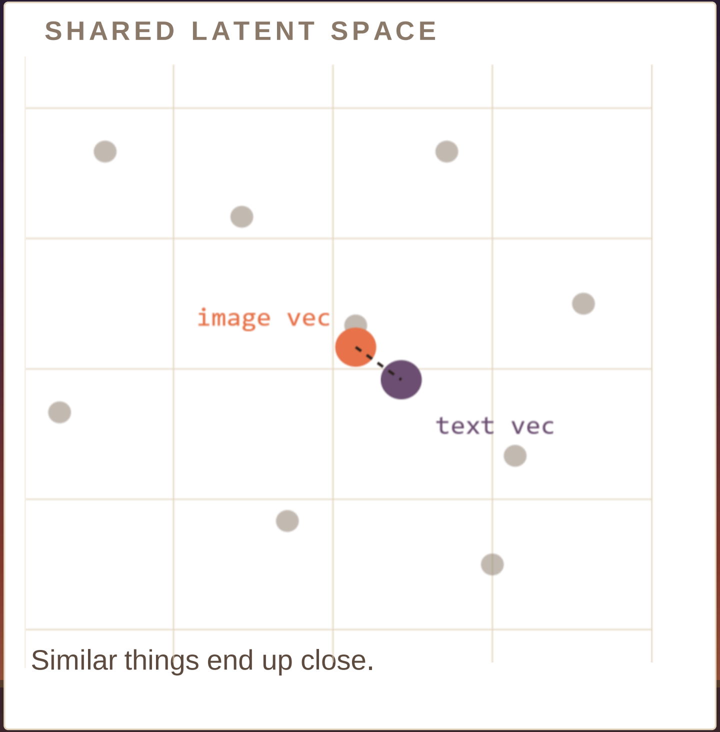
[0.21, -0.44, 0.87, -0.05, ..., 0.17]

512 vector dimension

CLIP

[0.20, -0.42, 0.85, -0.04, ..., 0.18]

512 vector dimension



CLIP: one shared space for images and text.

Search – FAISS Index Architecture

IndexFlatIP
How similarity is computed.

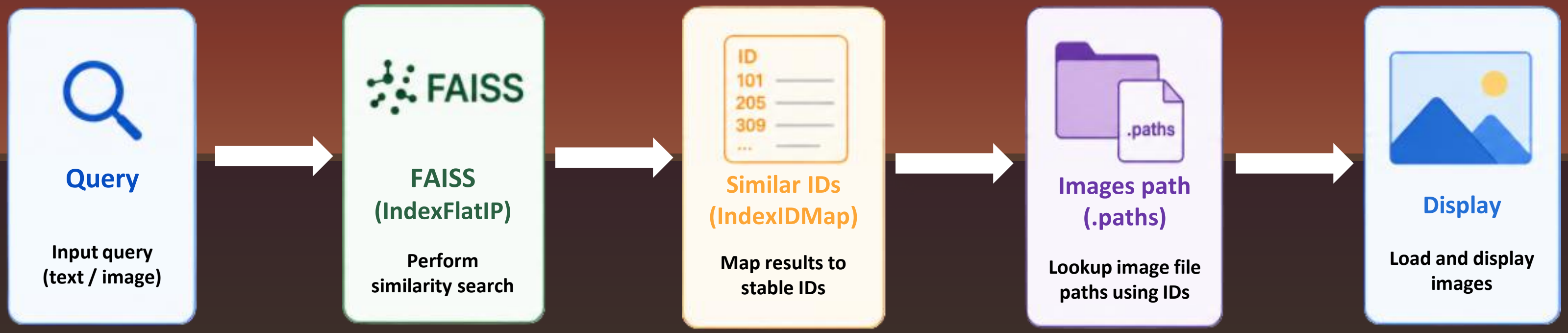
- Brute-force inner-product search
- L2-normalization → cosine similarity
- Exact Nearest Neighbor Search in milliseconds- no approximation

IndexIDMap
Which image a result points to.

- Wraps FAISS index to assign explicit 64-bit IDs per vector
- Add and delete vectors without shifting
- Path mapping survives mutation

.paths
The bridge to the filesystem.

- External lookup:
  ID → image path
- Synced with index after addition or deletion
- Prevents duplicate indexing



Cluster to Discover

Search finds what you ask for. Clustering shows you what you have.

The BERTopic Pipeline

Embed → Reduce → Cluster → Label

CLIP

One embedding per photo, same model as search.

UMAP

Preserve local neighborhood while lowering vector dimensions.

HDBSCAN

Density-based clustering.

BLIP + Ollama

Per-cluster caption and kw → local LLM rewrites it as a theme.

CLIP produces 512-dim vectors that capture visual semantics.

UMAP compresses vectors to a lower dimension.

HDBSCAN finds dense regions → clusters.

BLIP captions; Ollama turns them into a readable group label.

LIVE **DEMO**

“Pray with me” time

Key Takeaways

Local-first wins

Privacy, offline, zero cost — your data never leaves your machine

CLIP unifies modalities

One shared embedding space for images and text makes semantic search trivial

Quite simple stack, real power

FAISS + BERTopic = search + clustering

Built for your own data

No cloud API, no subscription — works with old photos, new photos, customizable as desired

What's Next

Temporal + EXIF search

Combine CLIP with image metadata (date, GPS, etc) to search for “photos from last Christmas”

File watcher auto-indexing

Watchdog monitors your photo folder and indexes new files automatically

Auto-generated photo stories

Feed cluster labels + representative photos to an LLM — get a narrative photo book

From Images to Audio

Automatically group music with same genre

Search for a particular song by mood or instruments

Thank you!

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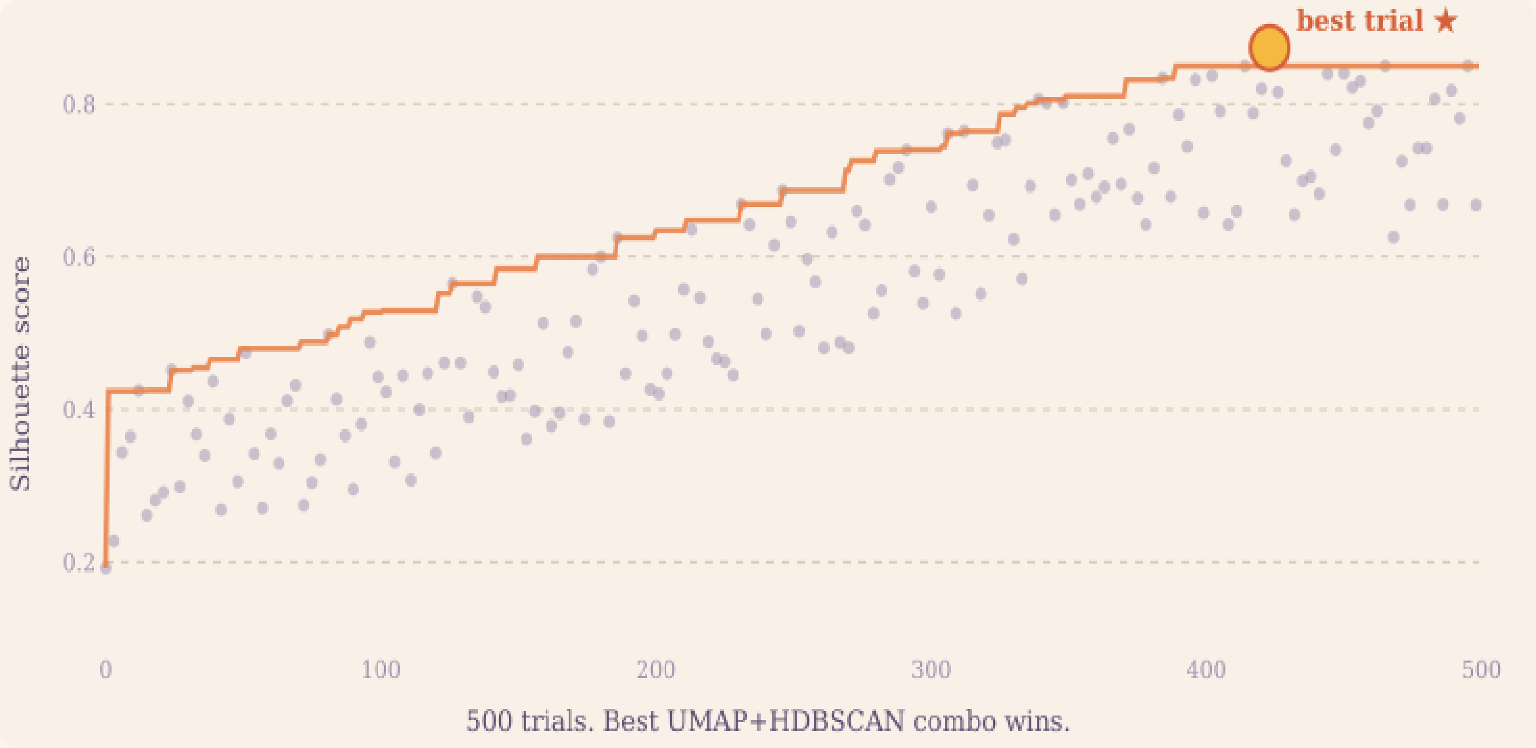
Optuna – Study and Find the Best Hyperparameters

UMAP and HDBSCAN have 8+ hyperparameters. Grid search would take days.



Index & Search - The Stack

FastAPI

OpenAI CLIP

FAISS

Streamlit

Clustering - The Stack

FastAPI

OpenAI CLIP

BERTopic

BLIP

OLLAMA

Streamlit